

Product Requirement Document (PRD) Template

Course / Track: On-the-Job Training (OJT) Project Framework

1. Document Overview & Metadata

Field	Description / Student Inputs
Project Title	[Insert Official Project Name]
Project Track	[Advanced Data Science / Data Science Research / Gen AI / Multimodal AI]
Team Members & IDs	[Member 1 Name & ID, Member 2 Name & ID]
Assigned Mentor	[Mentor Name]
Document Version & Date	v1.0 [Date]

2. Executive Summary & Problem Statement

2.1 Problem Statement

Provide a clear, concise statement of the problem your project addresses. Explain why this problem exists, who is affected by it, and why current solutions (if any) are insufficient.

2.2 Project Objectives

- Define the primary technical and functional goals of the system.
- Specify the key metrics for success (e.g., accuracy target, throughput, response latency).
- Detail the target audience and business or practical impact of the completed system.

3. System Architecture & High-Level Design

3.1 End-to-End System Pipeline

Describe the high-level workflow from user interaction/data input to the final response or model execution.

- **Data Ingestion:** Sources, streaming vs. batch processing, raw input formats.
- **Processing & Modeling Engine:** Core algorithms, model inference pipelines, and feature transformations.
- **Storage Layer:** Databases, vector indices, file systems, or object stores.
- **Presentation Layer:** User interface framework (Web/Mobile/API endpoint).

3.2 Architecture Diagram Placeholder

[Insert or attach a high-level flowchart/architecture diagram showing data flow and component integration.]

4. Technical Stack & Infrastructure

Component	Technologies / Frameworks Chosen	Rationale / Justification
Frontend / UI	e.g., React, Flutter, Streamlit, Gradio	[Why chosen]
Backend API	e.g., FastAPI, Flask, Node.js	[Why chosen]
ML / AI Frameworks	e.g., PyTorch, TensorFlow, Hugging Face, Scikit-Learn	[Why chosen]

Component	Technologies / Frameworks Chosen	Rationale / Justification
Database & Storage	e.g., PostgreSQL, MongoDB, Pinecone, FAISS	[Why chosen]
Deployment Platform	e.g., AWS, GCP, Vercel, Hugging Face Spaces	[Why chosen]

5. Functional & Non-Functional Requirements

5.1 Functional Requirements

1. **Feature 1 (Core Module):** Detailed description of user interaction, expected input, system behavior, and generated output.
2. **Feature 2 (API Layer):** Description of specific REST/gRPC endpoints and payloads.
3. **Feature 3 (Model Inferencing):** Real-time vs. batch processing capabilities.

5.2 Non-Functional Requirements

- **Performance & Latency:** Target response time per request (e.g., inference latency < 200ms).
- **Scalability & Concurrency:** Capability to handle simultaneous user sessions or large batch jobs.
- **Security & Privacy:** API key management, data encryption standards, and compliance rules.
- **Reliability & Availability:** Fault tolerance, graceful fallback handling, and error logging.

6. Data Strategy & Management

- **Dataset Source & Acquisition:** Specify public datasets, scraped data sources, or synthetic generation methods.
- **Data Preprocessing & Cleaning:** Plan for missing values, normalization, tokenization, or image augmentations.

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- **Data Privacy & Compliance:** Anonymization steps and user data consent handling.
 - **Data Version Control:** Versioning strategy for datasets and models (e.g., DVC, Git LFS).

7. Track-Specific Requirements

7.1 Track A: Advanced Data Science

- **Feature Engineering:** Statistical transformations, dimensionality reduction, feature selection methods.
- **Model Evaluation Suite:** Primary evaluation metrics (Precision, Recall, F1-Score, ROC-AUC, RMSE) and baseline comparison.
- **Analytics & Dashboarding:** Visualizing metrics, hypothesis testing outputs, and exploratory analysis interfaces.

7.2 Track B: Data Science Research Publications

- **Literature Review & Novelty Statement:** Summary of baseline papers and explicit identification of the research gap.
- **Methodological Novelty:** Mathematical formulations, algorithmic alterations, or novel loss functions.
- **Publication Plan:** Target venue, benchmarking methodologies, and comparative ablation studies.

7.3 Track C: Generative AI

- **Agent & Orchestration Architecture:** RAG pipelines, vector embeddings, RL environment/reward definition, or agent tool selection.
- **Fine-Tuning & Prompt Strategy:** Foundation model selection, PEFT/LoRA fine-tuning methodology, and prompt templates.
- **Safety & Benchmarking:** Hallucination mitigation techniques, evaluation metrics (e.g., ROUGE, BLEU, RAGAS), and fallback strategies.

7.4 Track D: Multimodal AI

- **Multimodal Data Fusion:** Early, late, or cross-attention fusion strategies for multi-stream inputs.
- **Model Architecture:** Encoder-decoder setups, vision-language transformers, or feature mapping models.
- **Explainable AI (XAI) & Latency:** Model interpretability methods (e.g., Grad-CAM, SHAP)

and real-time inference optimization.

8. Testing, QA & Validation Strategy

- **Unit & Integration Testing:** Automated code verification for data pipelines and API contracts.
- **Model Validation & Benchmarks:** Cross-validation strategies and resistance to data drift or adversarial inputs.
- **Edge Case Handling:** Graceful degradation strategy during unexpected inputs or network failures.

9. Deployment & CI/CD Strategy

- **Version Control Strategy:** Branching model (e.g., `main`, `development`, feature branches).
- **Containerization:** Dockerization strategy for reproducible builds.
- **CI/CD & Live Link:** Automated testing, build, and automated deployment trigger configuration.

10. Project Milestones & Evaluation Roadmap

Phase / Evaluation	Milestones & Deliverables	Deadline
PRD Approval Phase	Finalized PRD submitted, reviewed, and approved by Mentor.	30th Aug – 3rd Sept
GitHub Base Setup	Repository setup, architecture layout, baseline project structure.	6th September

Phase / Evaluation	Milestones & Deliverables	Deadline
Viva 1 (Month 1)	Data pipelines complete, baseline model implemented, initial API contracts set.	15th – 22nd Sept
Viva 2 (Month 2)	Core modeling/fine-tuning complete, UI integration, advanced feature validation.	15th – 22nd Oct
Viva 3 (Month 3)	Full end-to-end integration, performance optimization, initial deployment tests.	15th – 22nd Nov
Final Deliverables	Public live deployment link and project demonstration video submission.	25th November
External Final Viva	Comprehensive project defense, code review, and final demonstration.	15th – 22nd Dec

11. Risk Management & Constraints

Identified Risk	Impact / Severity	Mitigation Strategy
Compute / GPU resource limits	High	Use cloud free-tier GPUs (Google Colab/Kaggle), model

Identified Risk	Impact / Severity	Mitigation Strategy
		quantization, or sub-sampling.
API Rate Limits / Cost	Medium	Implement local caching, mock data layers, and rate-limiting wrappers.
Data Scarcity / Imbalance	Medium	Synthetic data augmentation, SMOTE, or transfer learning techniques.